

Executive Summary and Recommendations – Telecom Customer Churn Prediction

Project Overview

This project analyzes customer churn behavior in a telecom company using **Exploratory Data Analysis (EDA)** and **Machine Learning**. The dataset consists of **7,043 customer records** and **21 customer-related features**. The primary objective was to identify the factors influencing customer churn, build a predictive model to estimate churn probability, and generate actionable business recommendations to improve customer retention.

Data Preparation

The dataset was cleaned and preprocessed by handling missing values in the **TotalCharges** column, converting data types, and preparing categorical and numerical features for analysis. Exploratory Data Analysis was performed to understand customer behavior, after which the data was prepared for machine learning using **One-Hot Encoding**, **feature scaling**, and a **train-test split** before model training.

Exploratory Data Analysis (EDA)

The EDA revealed several important customer behavior patterns:

- Overall churn rate: **26.5%**
- Customers with **Month-to-Month contracts** exhibit the highest churn.
- **Fiber Optic** customers are more likely to churn.
- Customers with **shorter tenure** are at significantly higher risk.
- Customers without **Online Security** and **Tech Support** show increased churn.
- Higher **Monthly Charges** are associated with higher churn.
- **Electronic Check** users have the highest churn among payment methods.

These insights helped identify the major factors contributing to customer attrition.

Machine Learning Model

To move beyond descriptive analysis, a **Logistic Regression** model was developed using **Scikit-learn** to predict customer churn.

Data Preprocessing

- One-Hot Encoding for categorical features
- StandardScaler for numerical feature scaling
- 80-20 Train-Test Split

Model Evaluation

The model was evaluated using:

- Accuracy
- Precision
- Recall
- Classification Report
- Confusion Matrix

Feature coefficients were also analyzed to understand which variables contributed most to customer churn.

Major Churn Drivers

The strongest factors influencing churn include:

1. Month-to-Month Contract
 2. Low Customer Tenure
 3. Fiber Optic Internet Service
 4. Absence of Online Security
 5. Absence of Tech Support
 6. Electronic Check Payment Method
 7. Higher Monthly Charges
 8. Senior Citizen Status
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Business Recommendations

Based on the EDA findings and predictive modeling, the following actions are recommended:

1. Encourage customers to switch to long-term contracts through discounts and loyalty programs.
 2. Strengthen onboarding and retention initiatives during the first few months of service.
 3. Improve service quality and customer experience for Fiber Optic users.
 4. Bundle Online Security and Tech Support services with internet plans.
 5. Develop targeted retention campaigns for customers identified as high-risk by the predictive model.
 6. Provide personalized support for senior citizens and customers with high monthly charges.
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Conclusion

This project demonstrates an **end-to-end customer churn prediction pipeline**, beginning with data cleaning and exploratory analysis and extending to predictive machine learning. By combining business insights with a Logistic Regression model, the project identifies high-risk customer segments and provides data-driven recommendations that can help telecom companies reduce churn, improve customer satisfaction, and support long-term business growth.